Necessitism
Advanced Introduction to Metaphysics
15 November 2020

1. Background: unrestricted quantification

2. Necessitism, what it is, and what it isn't

Necessitism: $\Box\forall x\Box\exists y(y=x)$ (Necessarily, everything is necessarily something.)

- $\forall$ here is a *first-order* quantifier (though it's also very interesting to consider higher-order versions of necessitism).
- $\forall$ is *unrestricted*. Uncontroversially, $\Box\forall x(Fx \to \Box\exists y(Fy \wedge y=x))$ is false for many F.
- $\Box$ is metaphysical necessity (but note that opponents of necessitism typically also think it's false on much narrower interpretations of $\Box$)
- The initial $\Box$ is a bit of a red herring since opponents typically also reject $\forall$x$\Box\exists y(y=x)$ (call this NE).

3. Existence

If we stipulatively pronounce 'be identical to something' as 'exist', necessitism says 'Necessarily, everything necessarily exists'. But 'exists' already has a meaning, and necessitists may well argue that since some things could have failed to exist, existence requires more than just being something.

4. The "Direct Argument" for necessitism, and two kinds of contingentists

NI $\quad\Box\forall x\Box(x=x)$

IE $\quad\Box\forall x\Box(x=x \to \exists y(y=x))$

$\Box\forall x\Box\exists y(y=x)$

'Negative' contingentists accept the

Being Constraint $\Box\forall x_1\ldots\forall x_n\Box(Fx_1\ldots x_n \to \exists y(y=x_i))$

This means they accept IE and thus must reject NI.

By contrast, 'Positive' contingentists reject certain instances of the Being Constraint, presumably including IE.

5. Contingentist arguments by counterexample

'Necessarily, if my parents never met, nothing was me. But my parents could have never met. So, there could have been nothing was me.'

A counterfactual variant:

'If my parents had not met, nothing would have been me. But my parents could have never met. So, it could have been that nothing was me.'

Necessitists generally grant that if your parents hadn't met you wouldn't have been *concrete* (you would never have been born or conceived, never have had a shape or size or mass, never had any thoughts or experiences, …). They may also grant that you wouldn't have *existed*. They think the contingentists are conflating existential quantification restricted by some such property with genuinely unrestricted existential quantification.

Contingentists often deny that there could have been a non-concrete thing that could have been a concrete person, though this isn't strictly forced on them.

6. Necessitism and logical simplicity

Consider some plausible claims about metaphysical necessity.

$\Box\forall x(x{=}x)$

$\forall x\Box(Fx \rightarrow Fx)$

$\forall x\Box Fx \rightarrow \forall y(y = x \rightarrow Fy))$

Assembling lots and lots of such examples suggests a natural pattern: when a formula ϕ is a theorem of first-order logic, the *universal closure* of $\Box\phi$ is true. Since the formula $\exists y(y{=}x)$ is a theorem of first-order logic, $\forall x\Box\exists y(y{=}x)$ is an instance of this.

Contingentists must retreat to a weaker generalization: when ϕ is a theorem of *free* first-order logic, the universal closure of $\Box\phi$ is true.[1]

7. Entailment-theoretic argument for necessitism

To run these arguments we need to understand a (reflexive, transitive) *entailment* relation $\leq$ between properties. And for the arguments to be dialectically effective, we can't just define $F \leq G$ in terms of necessity and quantification, e.g. as $\Box\forall x(Fx \rightarrow Gx)$ or $\Box\forall x\Box(Fx \rightarrow Gx)$. Instead, we might define it in terms of necessity, e.g. as $F = \lambda x.Fx \wedge Gx$, or $G = \lambda x.Fx \vee Gx$, or maybe $\lambda x.Fx \wedge (Gx \vee \neg Gx) = \lambda x.Fx \wedge (Gx \wedge Gx)$…

We also need a notion of entailment $\leq$ among propositions, such that $F \leq G \rightarrow Fa \leq Ga$ and $p \leq q \rightarrow \lambda x.p \leq \lambda x.q$.

Adjunction: $(p \leq \forall F) \leftrightarrow (\lambda x.p \leq F)$[2]

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[1] Free logic:
PC: all tautologies
Free UI: $\forall x(\forall y\phi \rightarrow \phi[x/y])$
QK: $\forall x(\phi \rightarrow \psi) \rightarrow \forall x\phi \rightarrow \forall x\psi$
V: $\phi \rightarrow \forall x\phi$, x not free in ϕ
MP: $\phi \rightarrow \psi$, ϕ / ψ
GEN: ϕ / $\forall x\phi$

[2] Goodman uses a schematic version: $(\phi \leq \forall x\psi) \leftrightarrow (\lambda x.\phi \leq \psi)$, x not free in ϕ.

It may be helpful to break this up into three separate claims.[3]

Monotonicity: $F \leq G \rightarrow \forall F \leq \forall G$

Instantiation: $\lambda x.\forall F \leq F$

Vac: $p \leq \forall xp$

Instantiation is the only one that's needed to argue for individual instances of NE.

Argument: let E be $\lambda x.\exists y(y{=}x)$. By **Instantiation**, $\lambda x.\forall E \leq E$. So $(\lambda x.\forall E)a \leq Ea$ by **Application.** But $\forall E \leq (\lambda x.\forall E)a$ by **Abstraction**, so $\forall E \leq Ea$ by the transitivity of $\leq$, so $\Box(\forall E \rightarrow Ea)$. But $\Box\forall E$, so $\Box Ea$.

8. Contingentist responses

Negative contingentists reject Abstraction.

Positive contingentists reject Instantiation. They can keep Monotonicity and Vac, and replace Instantiation with the following[4]:

Commutativity: $(\forall F \wedge \forall G) \leq \forall x(Fx \wedge Gx)$

Distributivity: $\forall x(p \vee Fx) \leq (p \vee \forall F)$

Note that if $\forall$ has all these properties, so does its restriction $\lambda X.\forall y(Fy \rightarrow Xy)$ by any property F. So this package doesn't illuminate anything *distinctive* about universal quantification.

By contrast, if $\forall$ and $\forall^*$ both obey Adjunction, then for any F, $\forall F \leq \forall^* F$ and vice versa: Adjunction "fully characterizes" unrestricted quantification, up to mutual entailment.

9. Arguments for the abundance of contingent non-concrete

(*) 'There are six possible knives that could be made of these three handles and two blades'.

- 'Possible knife' means 'thing that is possibly a knife' (cf. 'every suspect is a possible murderer'). Don't hear it as 'knife that is possible'.

If (*) is true, it would still have been true if Blade 2 rather than Blade 1 had been attached to Handle 1. But then surely the knife that's in fact made of Blade 1 and Handle 1 would still have been one of the six possible knives; so it would have been something.

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[3] Given the reflexivity of $\leq$, Adjunction immediately implies Vac (taking F as $\lambda x.p$) and Instantiation (instantiating p as $\forall$F). For Monotonicity: suppose $F \leq G$. Then since $\lambda x.\forall F \leq F$ by Instantiation, $\lambda x.\forall F \leq G$ by the transitivity of $\leq$, so $\forall F \leq \forall G$ by Adjunction. To derive Adjunction from those three principles, suppose $p \leq \forall F$. Then $\lambda x.p \leq \lambda x.\forall F$ by the monotonicity of "padding"; and $\lambda x.\forall F \leq F$ by Instantiation, so $\lambda x.p \leq F$ by the transitivity of $\leq$. Conversely suppose $\lambda x.p \leq F$; then $\forall xp \leq \forall F$ by Monotonicity; but $p \leq \forall xp$ by Vacuity, so $p \leq \forall F$ by the transitivity of $\leq$.

[4] These are consequences of Adjunction, given that conjunctions are greatest lower bounds, disjunctions are least upper bounds, and negations an involution w.r.t. $\leq$. For Commutativity: since $\lambda x.\forall F \leq F$ and $\lambda x.\forall G \leq G$, $\lambda x.\forall F \wedge \forall G \leq \lambda x.Fx \wedge Gx$, which gives Commutativity by Adjunction. For Distributivity: $\neg p \leq \forall x \neg p$, so $\forall x(p \vee Fx) \wedge \neg p \leq \forall x(p \vee Fx) \wedge \forall x \neg p$, so $\forall x(p \vee Fx) \wedge \neg p \leq \forall x((p \vee Fx) \wedge \neg p)$ by Commutativity, so $\forall x(p \vee Fx) \wedge \neg p \leq \forall F$ by Monotonicity; so $\forall x(p \vee Fx) \leq p \vee \forall F$ by a Boolean manipulation. The other directions of Commutativity and Distributivity follow already from Monotonicity and Vac.

If contingentists conceded this, they would undermine their putative counterexamples to necessitism. So they have generally denied that (*) is (literally) true.

10. The contingentist paraphrase project

Many contingentists aren't comfortable just consigning the likes of (*) to the flames. They feel like that people who say things like (*) are sometimes "getting onto something true" even if their way of putting it can't be "taken at face value".

This prompts a quest for a "translation function" which will turn sentences like (*) into sentences not in tension with contingentism.

Fine's strategy: translate $\forall x Fx$ as $\exists p(p \wedge \text{WorldProp}(p) \wedge \Box\forall x\Box(p \rightarrow Fx))$.[5]

A simpler version: translate $\forall x Fx$ as $\exists p(p \wedge \Box\forall x\Box(p \rightarrow Fx))$

An entailment-theoretic variant: translate $\forall x Fx$ as $\exists p(p \wedge (\lambda x.p) \leq F)$.

11. Too successful?

Proponents of such translations typically accept that the translation of Necessitism is a truth.

Also, they think that the translation of $\forall x Fx$ entails $\forall x Fx$.

Also, the translations are systematic: it should be possible to implement them just by replacing each $\forall$ and $\exists$ with some complex higher-order predicate, leaving the structure of each sentence alone.

This raises a challenge: why isn't it the quantifier-meaning corresponding to the translation function, rather than the original one (which behaves logically like a restriction of it) that deserves the title 'unrestricted quantification'?

This is particularly pressing in the entailment-theoretic framework. If the putatively "non-face-value" quantifier-meanings are held to satisfy Adjunction, they enjoy a kind of logical distinction that makes it hard to understand why they don't count as 'quantifier-meanings' of which all others are 'restrictions'.

12. Weak necessitism

Here's a claim that these contingentists think necessitists are right about:

Weak Necesssitism: There is an Adjunctive property of properties (of objects).

Those who agree on this will find it useful to introduce some device for expressing such a property of properties, and for many purposes can bracket the question whether the label "unrestricted quantifier" applies to it.

Some higher-order contingentists may reject Weak Necessitism, but accept that there is a "Freely Adjunctive" property of properties of properties of properties such that Weak Necessitism comes out true when 'There is' is interpreted as expressing *that*. These 'weak weak necessitists' also have a way of stipulatively introducing a meaning of $\forall$ for which they'll accept Necessitism.

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[5] In his postscript to Prior's *Worlds, Times, and Selves.*